Understanding Distributed Cloud Computing, Edge Computing, Containers, Docker, Kubernetes, and DevOps

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Roll No. 04

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What is Distributed Cloud Computing?

- - Definition: A model where cloud services are distributed across different physical locations while remaining centrally managed.
- - Key Benefits:
- • Enhanced scalability
- • Improved performance and latency
- • Regulatory compliance
- - Use Cases: Multi-cloud strategies, disaster recovery, global applications.

What is Edge Computing?

- - Definition: A computing paradigm that brings computation and data storage closer to the location where it is needed.
- - Key Benefits:
- • Reduced latency
- • Lower bandwidth costs
- • Real-time processing
- - Use Cases: IoT, autonomous vehicles, industrial automation.

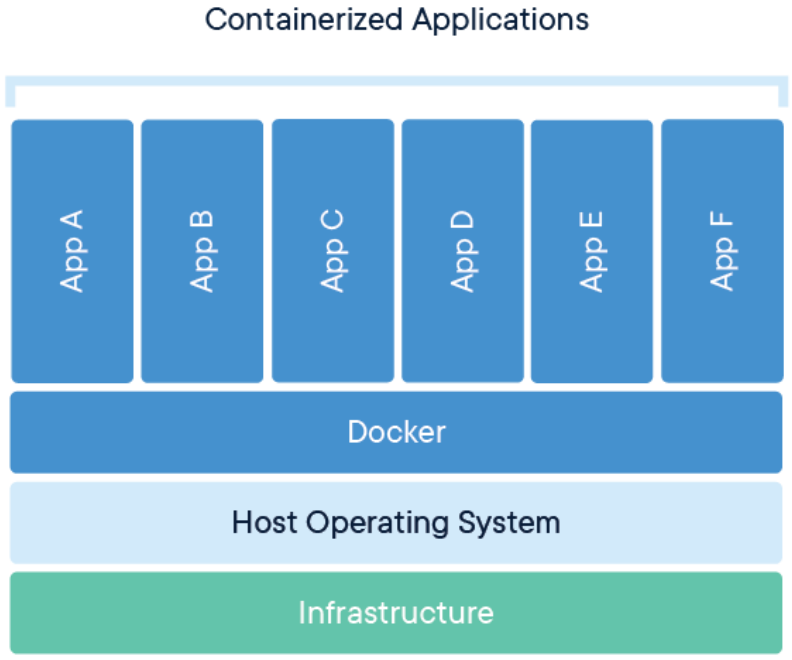
Distributed Cloud vs Edge Computing

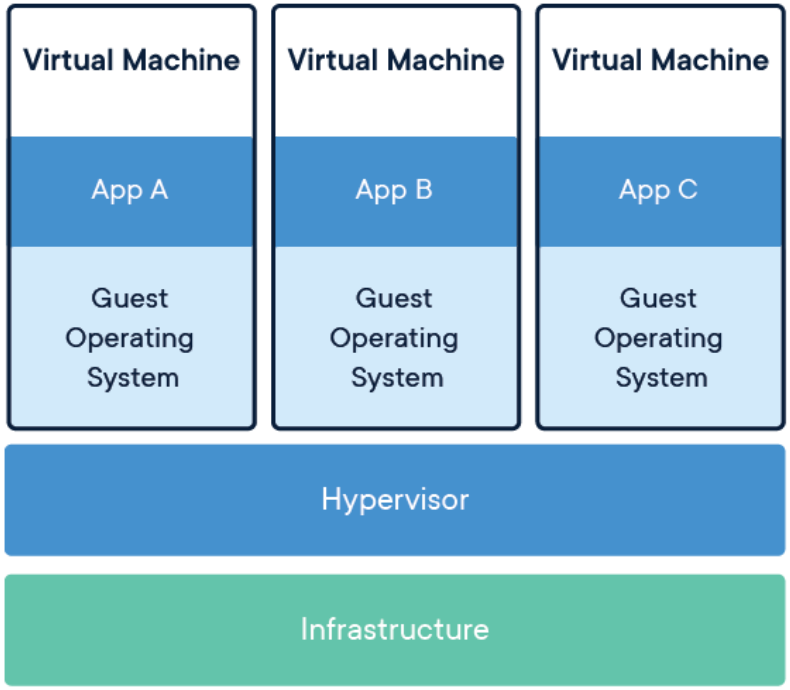
- - Distributed Cloud: Centralized management, scalable, used for multi-cloud services.
- - Edge Computing: Decentralized, ultra-low latency, used for real-time processing.

What are Containers?

- - Lightweight, portable, and isolated units that run applications and their dependencies.
- - Key Benefits:
- • Consistency across environments
- • Faster deployment
- • Scalability.

CONTAINERS

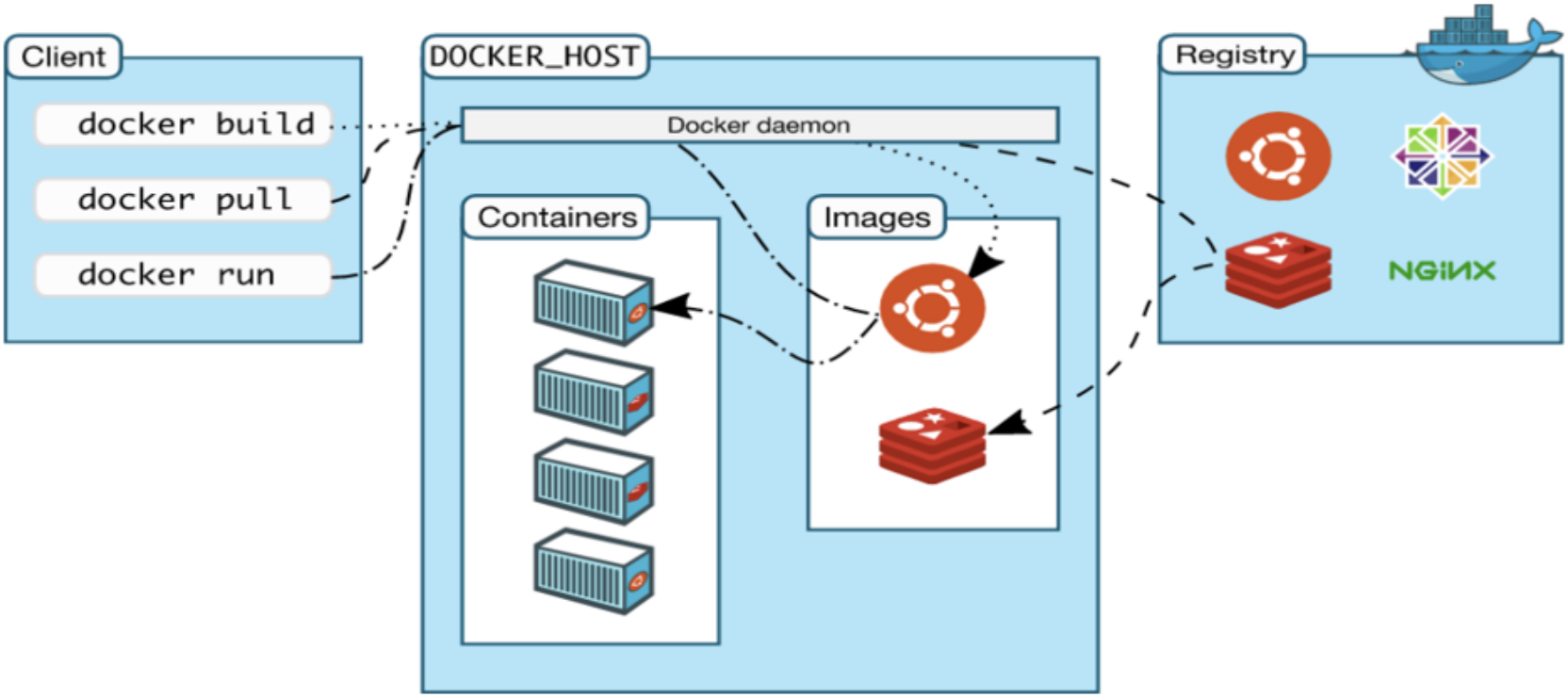

Containerized Applications
App A
App B
App C
App D
App E
App F
Docker
Host Operating System
Infrastructure


Virtual Machine
App A
Guest Operating System
Virtual Machine
App B
Guest Operating System
Virtual Machine
App C
Guest Operating System
Hypervisor
Infrastructure

Docker Overview

- - A platform for creating, deploying, and running applications in containers.
- - Why Docker?
- • Easy to set up and use
- • Environment consistency
- • Efficient resource utilization.

WORKING OF DOCKER



Kubernetes Overview

- - Open-source container orchestration platform for managing, scaling, and deploying containers.
- - Key Features:
- • Automated scaling
- • Self-healing capabilities
- • Load balancing.

Docker vs Kubernetes

- - Docker: Containerization, limited scalability, manual deployment.
- - Kubernetes: Orchestration, highly scalable, automated deployment.

What is DevOps?

- - A combination of practices and tools integrating software development and IT operations.
- - Core Principles:
- • Collaboration
- • Automation
- • Continuous integration and deployment (CI/CD).

DevOps Lifecycle

- 1. Plan: Define requirements and project scope.
- 2. Develop: Write and test code collaboratively.
- 3. Build: Package applications using CI/CD pipelines.
- 4. Test: Automated and manual testing.
- 5. Release & Deploy: Controlled application deployment.
- 6. Operate & Monitor: Continuous monitoring and improvement.

Summary

- - Distributed Cloud vs Edge Computing: Centralized vs decentralized.
- - Containers & Docker: Lightweight application deployment.
- - Kubernetes: Container orchestration.
- - DevOps: Streamlined software development and deployment.

THANK YOU

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